

Parse Guide Capture Protocol

- Proceed with the Parse Evercode Whole Transcriptome kit protocol until before cDNA amplification

At cDNA amplification step:

- Add guide additive PCR primer (10 μ M): 1 μ l in each 100 μ l sublibrary
- Use the following cycling conditions (modified from the original Parse protocol):

95°C for 3 min	10-18 cycles*
98°C for 20 sec	
65°C for 45 sec	
72°C for 3 min	
72°C for 5 min	
4°C hold	

* Number of cycles varies depending on the number of input cells, see Parse protocol

Post-amplification SPRI clean-up:

- Place the tubes with amplified cDNA against a magnetic rack (high setting) and wait for all the beads to bind to the magnet (~2 min: liquid should be clear). **CRITICAL!** Do NOT discard the supernatant.
- Transfer 90 μ l of the clear supernatant into new 100 μ l PCR tubes. Discard the original tubes with magnetic beads.
- Add 90 μ l of SPRI beads to each sublibrary. (1X cleanup, modified from the original Parse protocol to preserve shorter random hexamer guide products.)
- Pipette to mix 10 times with the pipette volume set to 150 μ l.
- Incubate at room temperature for 5 minutes.
- Place the tubes with amplified cDNA against a magnetic rack (high setting) and wait for all the beads to bind to the magnet (~2 min: liquid should be clear).
- With SPRI beads still against the magnetic rack, aspirate and discard the clear supernatant.
- Without resuspending beads, add 180 μ l of 85% ethanol and wait for 1 minute.
- Aspirate and discard the ethanol.
- Repeat the ethanol wash once.
- Remove tubes from the magnetic rack and centrifuge briefly (2-3 seconds).
- Place the tubes back on the rack (low setting) and use a P20 pipette to aspirate any residual ethanol.
- Air dry the beads (2 min) **CRITICAL!** Do NOT over-dry the beads. Over-drying of beads can lead to substantial losses in yield. “Cracking” of the beads is a sign of over-drying.
- Resuspend beads from each tube in 20 μ l (for Mini and Regular kits) or 25 μ l (for Mega kits) of molecular biology grade water
- Incubate the tubes at 37°C for 10 minutes to maximize elution of amplified cDNA.
- Place the tubes against the magnetic rack (low setting) and wait for all the beads to bind to the magnet (~2 min: liquid should be clear).

- Transfer 20 or 25 μ l of the elutant into new PCR tubes with a P20 pipette. Discard the tubes with SPRI beads. The amplified cDNA (and guide) library is now ready to be quantified.
- Measure concentration of the library using the Qubit dsDNA HS protocol.
- Run libraries on a 2% gel, BioAnalyzer, or tapestation to check if the expected size distribution is there.

Preparing cDNA libraries for sequencing:

- Follow the Parse protocol for fragmentation, adaptor ligation, and sample indexing.

Guide library preparation (gd PCR1):

- For every amplified cDNA library setup the following gd PCR1 reaction:

	1X
Amplified cDNA	20 ng
2X KAPA Hifi PCR Master Mix (Kapa Biosystems 07958935001)	25 μ l
Guide nested primer (10 μ M)	1.25 μ l
Guide index primer (10 μ M)	1.25 μ l
Water	Up to 50 μ l

- Run the following PCR program:

98°C for 3 min	
98°C for 20 sec	20-25 cycles*
60°C for 30 sec	
72°C for 30 sec	
72°C for 5 min	
4°C hold	

* Number of cycles varies depending on the number of input cells

- Expected product size: ~530 bp

Post gd PCR1 SPRI clean up:

- Add 50 μ l of SPRI beads to each guide library (1X cleanup) for a total volume of 100 μ l.
- Mix well by pipetting 10 times with the P200 pipette volume set to 75 μ l.
- Incubate at room temperature for 5 minutes.
- Place the tubes with amplified guides against a magnetic rack (high setting) and wait for all the beads to bind to the magnet (-2 min: liquid should be clear).
- With SPRI beads still against the magnetic rack, aspirate and discard the clear supernatant with a pipette.
- Without resuspending beads, add 180 μ l of 85% ethanol and wait for 1 minute.
- Using a pipette, aspirate and discard the ethanol from each tube.
- Without resuspending beads, add another 180 μ l of 85% ethanol and wait for 1 minute.

- Using a pipette, aspirate and discard the ethanol from each tube.
- Remove tubes from the magnetic rack and centrifuge quickly (2-3 seconds).
- Place tubes back on the rack (low setting) and with a P20 pipette remove any residual ethanol.
- With the tubes still on the rack, air dry the beads (~2 min). **CRITICAL!** Do NOT over-dry the beads. Over-drying of beads can lead to substantial losses in yield. “Cracking” of the beads is a sign of over-drying.
- Remove tubes from the magnet and resuspend beads in 20 µl of molecular biology grade water.
- Incubate at room temperature for 5 minutes.
- Place the tubes against the magnetic rack (low setting) and wait for all the beads to bind to the magnet (~2 min: liquid should be clear).
- Transfer 20 µl of the elutant into new PCR tubes with a P20 pipette. Discard the tubes with SPRI beads. The amplified guide library is now ready to be quantified.
- Measure concentration of the cDNA using the Oubit dsDNA HS protocol.
- Run libraries on 2% gel, BioAnalyzer or tapestation to check if the expected size distribution is there.

Guide library preparation (gd PCR2):

- For every amplified cDNA library setup the following gd PCR2 reaction:

	1X
gd PCR1 product	20 ng
2X KAPA Hifi PCR Master Mix (Kapa Biosystems 07958935001)	25 µl
P7 primer (10 µM)	1.25 µl
SI-PCR primer (10 µM)	1.25 µl
Water	Up to 50 µl

- Run the following PCR program:

98°C for 3 min	5-7 cycles*
98°C for 20 sec	
60°C for 30 sec	
72°C for 30 sec	
72°C for 5 min	
4°C hold	

* Number of cycles varies depending on the number of input cells

- Expected product size: ~575 bp

Post gd PCR2 SPRI clean up:

- Add 50 µl of SPRI beads to each guide library (1X cleanup) for a total volume of 100 µl.
- Mix well by pipetting 10 times with the P200 pipette volume set to 75 µl.
- Incubate at room temperature for 5 minutes.

- Place the tubes with amplified guides against a magnetic rack (high setting) and wait for all the beads to bind to the magnet (~2 min: liquid should be clear).
- With SPRI beads still against the magnetic rack, aspirate and discard the clear supernatant with a pipette.
- Without resuspending beads, add 180 µl of 85% ethanol and wait for 1 minute.
- Using a pipette, aspirate and discard the ethanol from each tube.
- Without resuspending beads, add another 180 µl of 85% ethanol and wait for 1 minute.
- Using a pipette, aspirate and discard the ethanol from each tube.
- Remove tubes from the magnetic rack and centrifuge quickly (2-3 seconds).
- Place tubes back on the rack (low setting) and with a P20 pipette remove any residual ethanol.
- With the tubes still on the rack, air dry the beads (~2 min). **CRITICAL!** Do NOT over-dry the beads. Over-drying of beads can lead to substantial losses in yield. “Cracking” of the beads is a sign of over-drying.
- Remove tubes from the magnet and resuspend beads in 20 µl of molecular biology grade water.
- Incubate at room temperature for 5 minutes.
- Place the tubes against the magnetic rack (low setting) and wait for all the beads to bind to the magnet (~2 min: liquid should be clear).
- Transfer 20 µl of the elutant into new PCR tubes with a P20 pipette. Discard the tubes with SPRI beads. The amplified guide library is now ready to be quantified.
- Measure concentration of the cDNA using the Oubit dsDNA HS protocol.
- Run libraries on 2% gel, BioAnalyzer or tapestation to check if the expected size distribution is there.

Oligonucleotide sequences (5' -> 3'):

- Guide additive PCR primer:
 - GAGGGCCTATTTCCCATGATT*C*C
- Guide nested PCR primer
 - CTACACGACGCTCTTCCGATCTGTGGAAAGGACGAAACACC
- Guide index primer (P7-i7-R2):
 - CAAGCAGAAGACGGCATACTGAGATATGTCCGTGTGACTGGAGTTCAGACGTGTGC
 - Use a different i7 index for each sublibrary
- 10X Genomics SI-PCR primer:
 - AATGATACGGCGACCAACGAGATCTACACTCTTCCCTACACGACGC*T*C
- P7 primer:
 - CAAGCAGAAGACGGCATACTG